

Flying Liferaft

**Initial Wing Planform Design, Nose design and Star CCM+
analysis on the Aerodynamic Performance of Fregata**



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Abstract

This report outlines writers contribution in the aerodynamic design of Fregata, a small sized unmanned aerial vehicle (UAV) which can deliver immediate buoyancy supports to drowning personnel in the sea. The results are also presented in this report. The design process was started by an initial wing planform design based on design requirements and values from the previous stages. The initial wing design had an area of 0.67m^2 , span of 2m, taper ratio of 0.5 and twist of -2.5° , and made up of NACA 4412 airfoil section. Following was a study on the nose design. Working closely with the structural team, it was designed to have an elliptical nose with length of 0.47 m and base radius of 0.176 m. Finally, Star CCM+ analysis was done to validate the designs and give accurate aerodynamic performance data. It was also able to spot out some design flaws and driven the redesign of a more capable wing and the addition of aerodynamic fairings on the VTOL motors.

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List of Symbols

C_L	Lift coefficient
$C_{L_{des}}$	Design lift coefficient
$C_{L_{max}}$	Maximum lift coefficient
C_D	Drag coefficient
C_{D_i}	Induced drag coefficient
W/S	Wing loading (N/m ²)
b	Span (m)
$\wedge$	Dihedral (deg)
a	Lift curve slope (rad ⁻¹)
α_0	Zero Lift AoA (deg)
λ	Taper ratio
S_{ref}	Reference Area (m ²)
AR	Aspect Ratio
L/D	Lift to drag ratio
L/D_i	Lift to induced drag ratio

1 Introduction

“Fregata” is a small sized unmanned aerial vehicle (UAV) which can deliver immediate buoyancy supports to drowning personnel in the sea. It has vertically takeoff and landing (VTOL) capability, is able to cruise at a speed of 58.3ms^{-1} and covers an operation radius of 70km. Although Fregata has the characteristics of both a fixed wing propeller aircraft and quadcopter, the aerodynamic team mainly focuses on the wing, empennage and fuselage aerodynamics. The design process of Fregata is summarised in Figure 1 in chronological order and the topics highlighted in red are covered in this individual report.

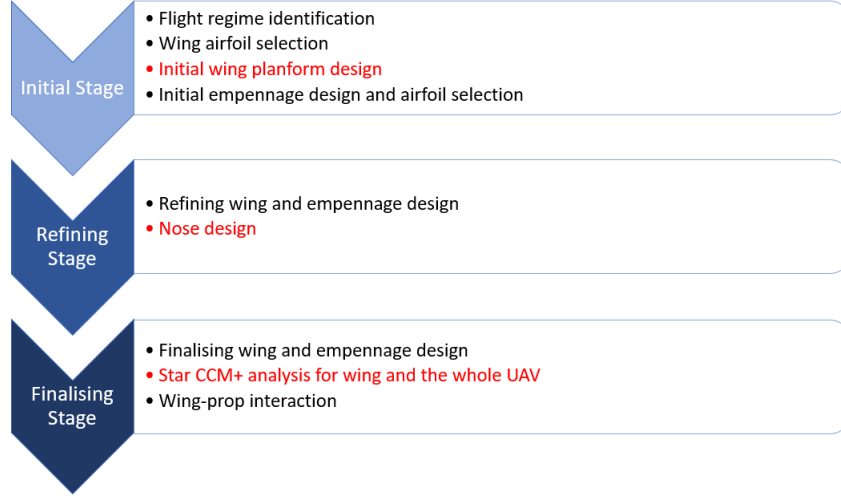


Figure 1: Design Timeline

The aerodynamic design of Fregata was started by generating wing geometry based on initial sizing values and design constraints, and utilising lifting line theory to predict the performance while optimising twist and taper ratio for the best lift and drag performance. A Matlab code was deployed to automate the process. The advantages of this approach were being able to generate the wing geometry quickly and estimate the aerodynamic performance of the 3D wing fairly accurately at an early stage of design. This was favourable for other teams to initiate their analysis.

Proceeding to the intermediate stage, a nose design study was carried out. Different nose shapes were studied using computational analysis to choose the nose shape that had the least drag.

During the last design stage, CFD analyses were done for the wing and fuselage using Star CCM+. We were able to get more accurate aerodynamics performance values and check if the design met the requirements.

2 Initial Wing Planform Design

2.1 Aims and visions

The wing will mainly be used in the cruise segment (at 58.3 ms^{-1}) to generate enough lift for steady level flight. Therefore, the design lift coefficient $C_{L_{des}}$ is 0.35 and the optimisation goal is to minimise the drag at this $C_{L_{des}}$. Also, the wing should be capable to attain high enough $C_{L_{max}}$ to ensure a stall speed of at most 30 ms^{-1} which is requested by control team for smooth transition.

The starting point of the wing planform design are a couple of parameters chosen in previous stages and some constraints imposed by the overall design considerations, which are listed in Table 1 below. Noted that the following values are not final because they were refined later in the intermediate

stage. In the following section, we would start from these values to outline the procedure of the initial wing planform design, and give prediction of lift and drag performance.

Table 1: Initial Wing Design Considerations

Parameter	Value	Justification
Wing loading W/S	832 N/m^2	Chosen from constraints diagram in the initial sizing stage
Span b	2 m	Decided due to storage and packing consideration
Sweep Λ	0°	Low Mach number $M = 0.17$, no need for sweep
Airfoil:	NACA 4412	Chosen in the airfoil selection stage due to high L/D at $C_{L_{des}}$ [1]
Lift curve slope a	6.27	
Zero lift angle of attack α_0	-4.4°	
Minimum twist	-2.5°	Chosen to ensure a "washout" condition in which the wing stalls at the root before the tip, improving control during stalling [2]
Taper ratio λ	0.5 or 1	Requested by the control team due to more stability data available for these values of taper ratio

2.2 Theory and Design Procedures

The reference area S_{ref} of the wing is simply the weight of the UAV (in Newtons) divided by the chosen wing loading as shown in Table 1, which gives 0.61 m^2 . The aspect ratio AR could be calculated by b^2/S_{ref} , giving a value of 6.59.

2.2.1 Optimizing Goal

The missing parameters were then taper ratio λ and twist. It was known that 3D wing has reduced lift than the 2D airfoil due to trailing vortices [3]. And it was known that a wing with elliptical lift distribution was the most efficient, and more importantly taper ratio and twist could help achieving that for a 3D wing [2]. Therefore the next step was to determine the taper ratio and twist which gave the most efficient wing, in particular the one with maximum lift to induced drag ratio L/D_i at our $C_{L_{des}}$.

2.2.2 Prandtl Lifting line Theory

The Prandtl Lifting Line Theory was used to find what was the optimal combination of taper ratio and twist. It was able to predict the lift coefficient C_L and induced drag coefficient C_{D_i} of a 3D wing based on the geometry of the wing, and account for the effect of trailing vortices [3]. Since the UAV would be flying at a low Mach number of 0.17, compressibility effect is negligible. Together with the design decision of zero sweep allowed lifting line theory to give fairly accurate estimation of lift and induced drag [3]. The aerodynamic team understood that lifting line theory could not find the drag due to viscous effect, and more analysis was needed. However, considering the time needed to model that and the urgent needs of other teams to have initial values, lifting line theory was used as a starting tool.

For some arbitrary values of taper ratio and twist, the chord distribution c , in meters, and the effective angle of attack distribution α_e , in degree, of airfoil sections across the semi-span in polar coordinate (with $\theta = 0$ was at the wing tip and $\theta = \pi/2$ was at the centre line of the wing) were described by the following equations:

$$c(\theta) = c_r[1 - (1 - \lambda)|\cos \theta|] \quad (1)$$

where c_r was the chord in meters at the root, which could be calculated by:

$$c_r = \frac{2S_{ref}}{b(1 + \lambda)} \quad (2)$$

using the previously stated values of b and S_{ref} , and

$$\alpha_e(\theta) = \alpha_{inf} + \alpha_r + twist \times |\cos \theta| - \alpha_0 \quad (3)$$

where α_{inf} was the freestream angle of attack, setted as 0° in steady level flight, α_r was the root setting angle in degree, and α_0 was the airfoil's zero lift angle of attack listed in Table 1. Noted that in this initial stage, an linear distribution of taper ratio and twist was assumed.

The lifting line theory stated that:

$$\sum_{n=1}^{\infty} A_n \sin(n\theta)(n\mu + \sin\theta) = \mu \alpha_e \sin\theta \quad (4)$$

where α_e could be calculated by Equation 3 and μ is a function of θ :

$$\mu(\theta) = \frac{ac}{8S_{ref}} \quad (5)$$

with previous values of a and S_{ref} , and c to be calculated by Equation 2.

The C_L and C_{Di} of the wing could be calculated by:

$$C_L = \pi A R A_1 \quad (6)$$

$$C_{Di} = \pi A R \sum_{n=1}^{\infty} n A_n^2 \quad (7)$$

2.2.3 Iterative Matlab Code

To find the best combination of the taper ratio and twist, an iterative Matlab code was deployed to find the C_L and C_{Di} for all combinations of taper ratio and twist in our defined range, then found the one with the highest L/D_i . It utilised 6 collocation points for the lifting line equations which was later proven to be adequate for the validity of the results. The code is attached in Appendix A, but a flow chart of the code is shown below:

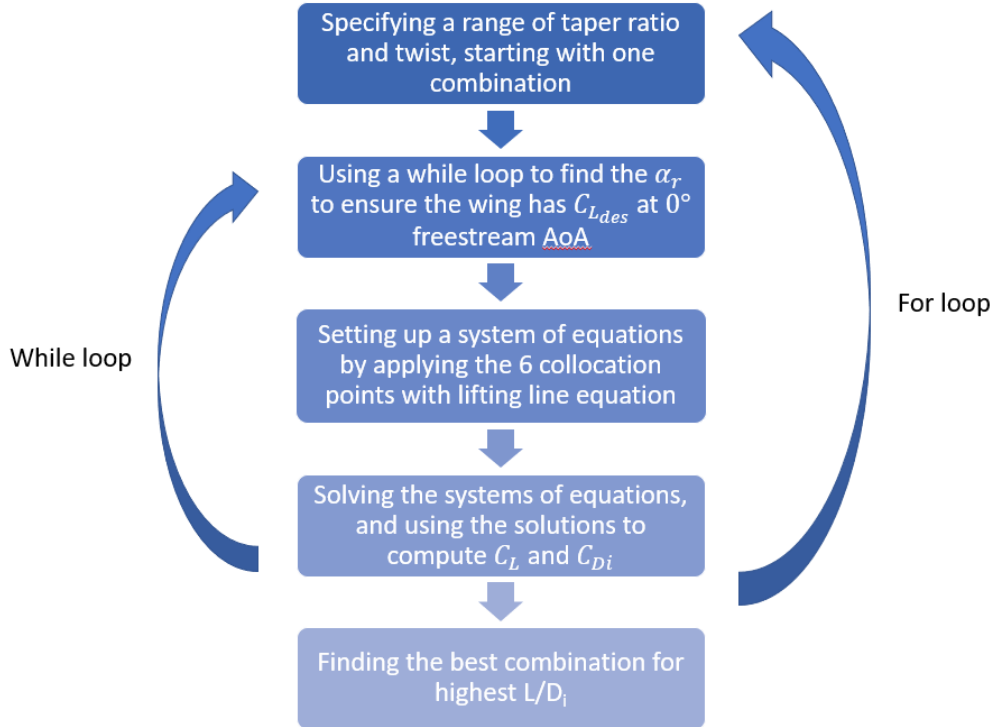


Figure 2: Matlab Code Flow Chart

The taper ratio was set to be either 0.5 or 1 as mentioned in Table 1. The twist was set between -2.5° and -5° such that the twist was high enough to ensure the root to stall before the tip and low enough for structural considerations.

2.2.4 Result and Validation

Using the methodology outlined above, it showed that the best combination of taper and twist being $\lambda = 0.5$ and twist $= -2.5^\circ$. The wing had a L/D_i of 42.5. The parameters of generated wing planform are shown in Table 2.

Table 2: Initial Wing Planform Parameters

S_{ref} (m ²)	b (m)	AR	c_r (m)	α_r (deg)	λ	Twist (deg)
0.61	2	6.59	0.405	1.8	0.5	-2.5

In the later stage, more in-depth analyses and optimisations were done using panel method by the Athena Vortex Lattice (AVL) software [1]. A comparison of the values of induced drag calculated by both methods were listed in Table 3. It showed that the lifting line code gave accurate estimation of the induced drag of the wing planform.

Table 3: Comparison of C_{D_i} by Lifting Line Code and AVL Panel Method

Lifting line code	0.0121
AVL panel method	0.0122

2.3 Conclusion

By utilising lifting line theory, an initial wing planform with parameters shown in Table 1 was designed and its performance was estimated in a fast and accurate manner. The iterative Matlab code has also served as quick tool to calculate lift and drag of the wing whenever there was a design updates. It has enabled the wing design process to be efficient.

3 Nose Design

3.1 Aims and visions

The overall fuselage of the UAV was blunt compared to the wing we have. Therefore it was expected that the fuselage would be accountable for the major source of drag. In order to minimise the drag of the whole aircraft, which could help saving fuel and cruise at a higher speed, the aerodynamic team had looked into finding the nose shape which gave the least drag.

3.2 Nose Shape Selection

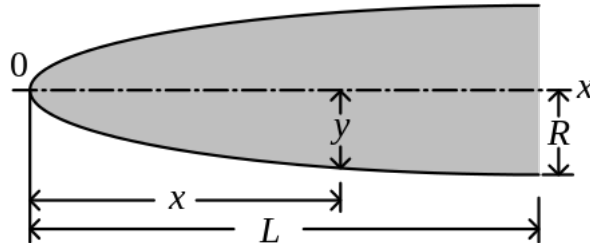


Figure 3: General Nose Parameters

Figure 3 shows the two major parameters of the nose. At the start of this stage, the structural team estimated the radius of the fuselage R to be 0.18m and the length of the nose L to be 0.345 m. A list of shortlisted possible nose shapes are shown in Table 4. These shapes are shortlisted because

of their curved geometries which are more suitable than sharp geometries in subsonic flight.

Table 4: List of Possible Nose Shapes

Nose Shape	Geometry equation [4]
Elliptical	$y = R\sqrt{1 - \frac{x^2}{L^2}}$
Tangent ogive	$y = \sqrt{\rho^2 - (L - x)^2} + R - \rho$ with $\rho = \frac{R^2 + L^2}{2R}$
Parabolic	$y = R\frac{2(x/L) - K'(x/L)^2}{2 - K'}$ for $0 \leq K' \leq 1$
Power series	$y = R(\frac{x}{L})^n$ for $0 \leq n \leq 1$
Hacck series	$\theta = \arccos(1 - \frac{2x}{L})$, and $y = \frac{R}{\sqrt{\pi}}\sqrt{\theta - \frac{\sin 2\theta}{2}} + C \sin^3 \theta$, for $0 \leq C \leq 2/3$

The drag of the shortlisted noses were investigated using Star CCM+. The model was set up to simulate the flight at cruise, with incoming air flowing at 58.3 ms^{-1} and a density of 1.21 kgm^{-3} . A laminar model was assumed because it was assumed that the flow would stay attached to the nose during forward flight. A $3\text{m} \times 5\text{m}$ rectangular domain was set up. The mesh was set up with the parameters shown in Table 5, and a close up mesh scene of an example nose is shown in Figure 4.

Table 5: Mesh Parameters: Nose Study

Mesh type	Polygonal mesh with prism layers on the nose
Base size	0.02 m
Prism layers number and thickness	10 and 4E-04 m
Custom control	Surface size of 0.001 m around the nose and its front

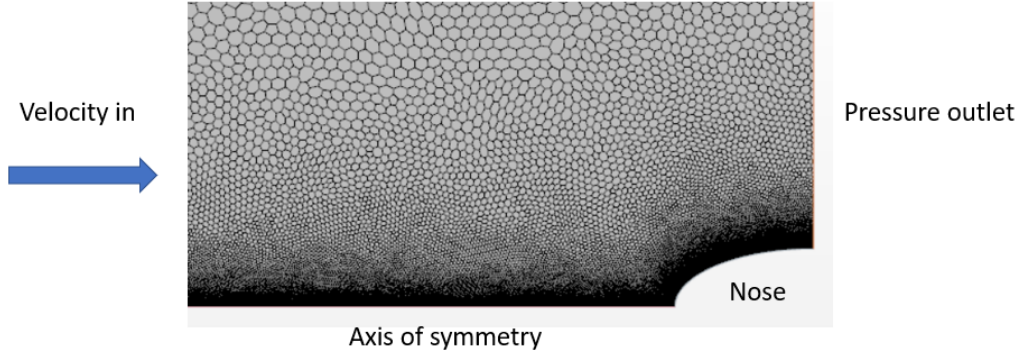


Figure 4: Close Up Mesh Scene of an Elliptical Nose

Since the UAV was designed to have circular fuselage, the simulation was done in 2D manner using “axis-symmetrical” condition to save time. The nose wall would revolve about the axis of symmetry as shown in Figure 4, creating a circular nose which joined well with the fuselage. The results are shown in Table 6, noted that the C_D were calculated based on S_{ref} .

As shown in Table 6, both elliptical nose and Hacck series nose with $C=2/3$ had a C_D of 0.001. A decision was made to go for an elliptical nose after a discussion with the structural team because it would be easier for them to model during structural analysis.

Table 6: C_D Values of Different Nose Shapes

Nose Shape	C_D
Elliptical	0.0010
Tangent ogive	0.0012
Parabolic $K'=1/2$	0.0031
Parabolic $K'=3/4$	0.0023
Parabolic $K'=1$	0.0014
Power series $n=1/2$	0.0022
Power series $n=3/4$	0.0033
Hacck series $C=0$	0.0017
Hacck series $C=1/3$	0.0013
Hacck series $C=2/3$	0.0010

3.3 Effect of nose length

A nose with higher fineness ratio (L/R) had less drag [2]. While R was fixed to the dimension of the main fuselage, there was a option to increase the L for less drag. This argument was supported by further Star CCM+ analysis of elliptical with different length, and the results are shown in Figure 5.

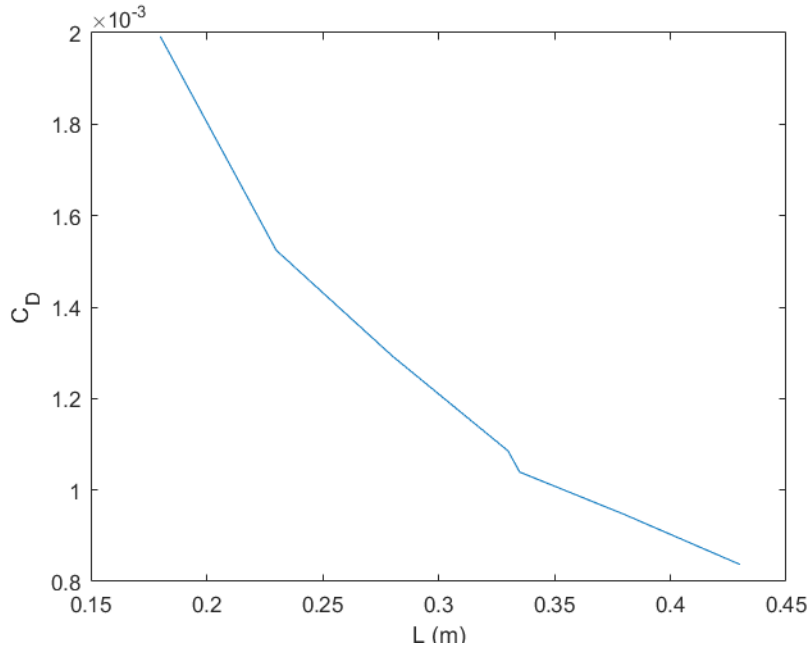


Figure 5: C_D of Elliptical Nose with Varying Length

Although the length of the nose was limited by different factors, such as internal system packing and wing placing position, a consensus was reached between the Aerodynamic team and the Structural team to increase the length of nose as long as possible.

3.4 Final Design

The final design of the nose was listed in the table below. It was placed right before the wing for maximum length.

Table 7: Final Nose Design

Nose Shape	Elliptical
Radius R	0.176 m
Length L	0.47 m

4 Star CCM+ Analysis on Aerodynamic Performance

The last stage of the aerodynamic design process was doing computational analysis to validate our design and produce accurate lift and drag data for other teams to refine their models. 3D models were imported into Star CCM+ for analysis. Noted that since it was time consuming to set up the models and run the simulations, the simulations were sometimes lagging behind the latest design. Nonetheless it still provided great insight into the aerodynamic performances of the whole aircraft.

4.1 Wing

Star CCM+ was used to validate if the wing design satisfies the design requirements. 3D wing models were imported to the Star CCM+ and tested for the lift and drag coefficient at different angle of attack.

4.1.1 Model Setup

Although different wing geometries were tested, they shared the same size of domain, physics model, flow condition and mesh parameters. The domain of the simulations was a 50m radius hemisphere with the test object on the centre. The physics model used in the simulations is 3D K-Omega turbulent model with coupled flow, constant air density. The flow conditions and mesh parameters were listed in Table 8 and 9.

Table 8: Flow Conditions

Reynold number	$Re = 1.3E+06$
Reference pressure	$P = 10E+05 \text{ Pa}$
Air density	$\rho = 1.21 \text{ kgm}^{-3}$
Flow speed	$u = 58.3 \text{ ms}^{-1}, M = 0.17$

Table 9: Mesh Parameters

Base size	0.05m - *8% around the lifting surface edges - *40% on the wake -50m on the domain boundary (same as domain radius)
Prism layer	30 layers and 0.009m total thickness by estimation of boundary layer thickness
Surface grow rate	“Slow” option

4.1.2 Clean Wing

The following wing planform with parameters shown in Table 10 was first tested. The design is the first version of optimised wing planform based on AVL results [1].

The results of the C_L of wing version 1 were plotted in Figure 6. It showed that the wing has met our requirement at low angle of attack. However, it stalled at a lower C_{Lmax} of 1.25 than expected which should be 1.45. It therefore gave a higher stall speed which was unfavourable for transition

Table 10: Wing Planform Parameters: Version 1

Airfoil	S_{ref} (m ²)	b (m)	AR	c_r (m)	α_r (deg)	λ	Twist (deg)
NACA 4413	0.67	2	6	0.444	1.4	0.5	-2.5

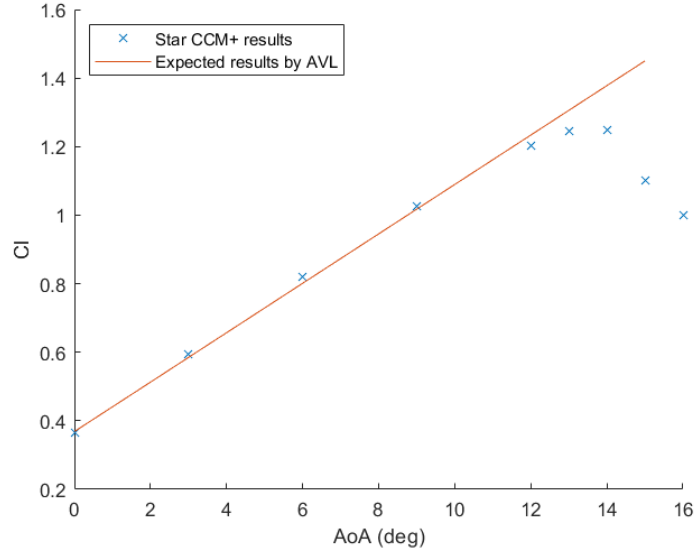


Figure 6: C_L Plot of Wing Version 1

from VTOL to forward flight. In this regard, modifications were made to the main planform, and the new planform parameters were listed in Table 11.

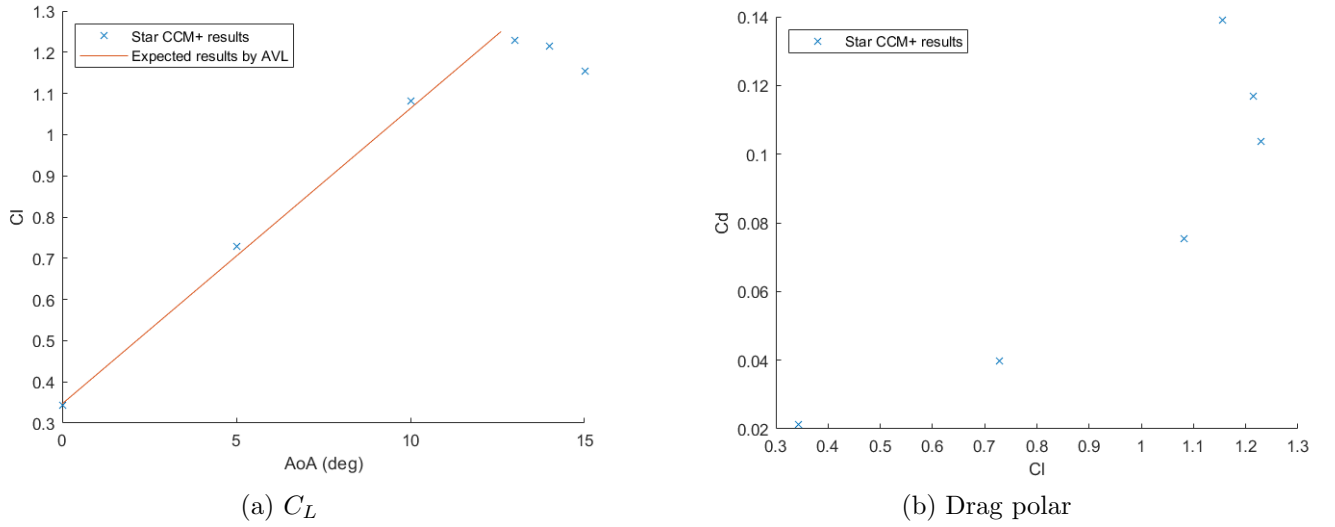


Figure 7: C_L Plot and Drag Polar of Wing Version 2

As shown in Figure 7, version 2 of the wing planform showed an agreement of between the desired lift performance and the Star CCM+ results. Although the C_{Lmax} of version 2 is lower than version 1's. The increased reference area allowed the UAV to stall at 30 ms^{-1} which was desirable for transition. Therefore this version was our final wing design. The Star CCM+ results showed that it had an Oswald efficiency of 0.8.

Table 11: Wing Planform Parameters: Version 2

Airfoil	S_{ref} (m ²)	b (m)	AR	c_r (m)	α_r (deg)	λ	Twist (deg)
NACA 4413	0.71	2.2	6.83	0.444	1.3	0.45	-4

4.1.3 Wing with VTOL motors

In the actual design, cylindrical VTOL motors were mounted pylons which were extended from the leading edge and trailing edge of the wing. Simulations were done to investigate the effect of the motor mounts to the aerodynamic performance. Noted that the simulations were done earlier than version 2 of wing was produced. Therefore, only qualitative and rough quantitative results are presented below.

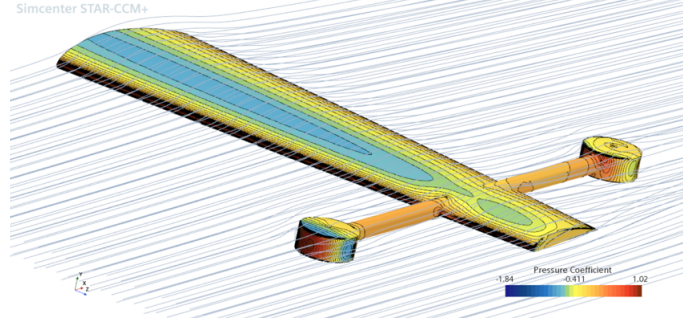


Figure 8: Flow Around the Motor Mount

Figure 8 showed that the wake of the motor mount has cut up the C_p contour on the wing surface. It led to a reduction in lift of about 5% and an increase in C_D of around 0.01.

To avoid the wake of the VTOL motors affecting the designed lift and drag characteristic of the wing, aerodynamic fairings were added to the motors. They were designed to have an ellipsoidal shape to help flow staying attached, and have the same frontal area as the motors to minimise drag. The overall length of the fairing has also taken the considerations of having minimum effect on the efficiency of the VTOL propellers. Figure 9 showed that the fairings were able to prevent the wake from affecting the lifting surface.

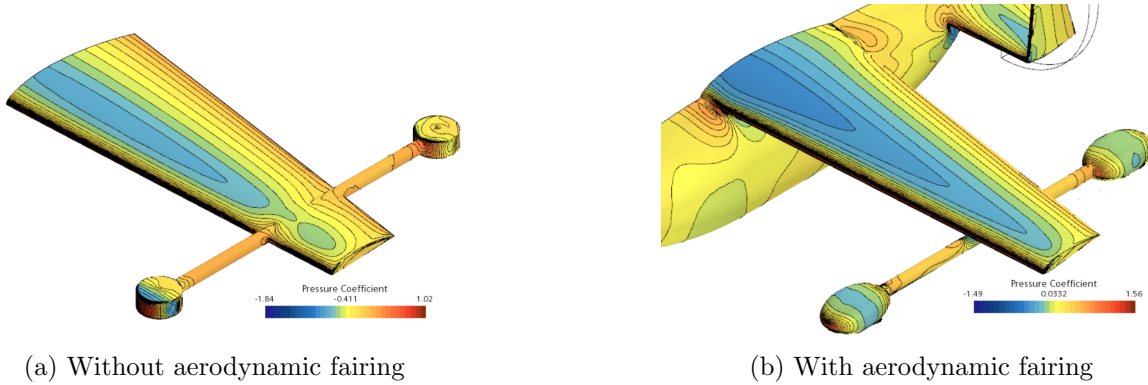


Figure 9: Comparison between the Pressure Coefficient Distribution of the Wing

4.2 Whole UAV

A Star CCM+ analysis was done on the UAV when all teams had their initial designs, as shown in Figure 10. Noted it was slightly different from the finalised design, as shown on the cover page, in terms of the VTOL propellers placement. The Star CCM+ analysis was done based on the version with propellers mounted below the motors. Despite the difference, it could be reasonably assumed

that both versions have similar aerodynamic performance at cruise because they were both locked in forward flight to align with the flow direction, thus having minimal effect on other surface.

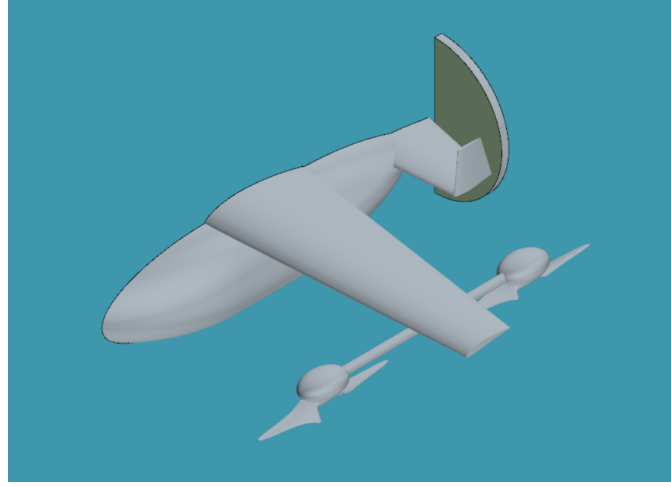


Figure 10: UAV Geometry for Star CCM+ Analysis

4.2.1 Model Setup

The simulation models were exactly the same as those during wing analysis, as described in Section 4.1.1. One added difference was that an actuator disk was added behind the UAV to model the effect of the pusher propeller. The actuator disk was placed at the same position and had the same size as the propeller. It would suck air in at a speed of 61.9 ms^{-1} and push air out at 65.5 ms^{-1} to provide the thrust needed at cruise. A resulting mesh scene is shown in Figure 11.

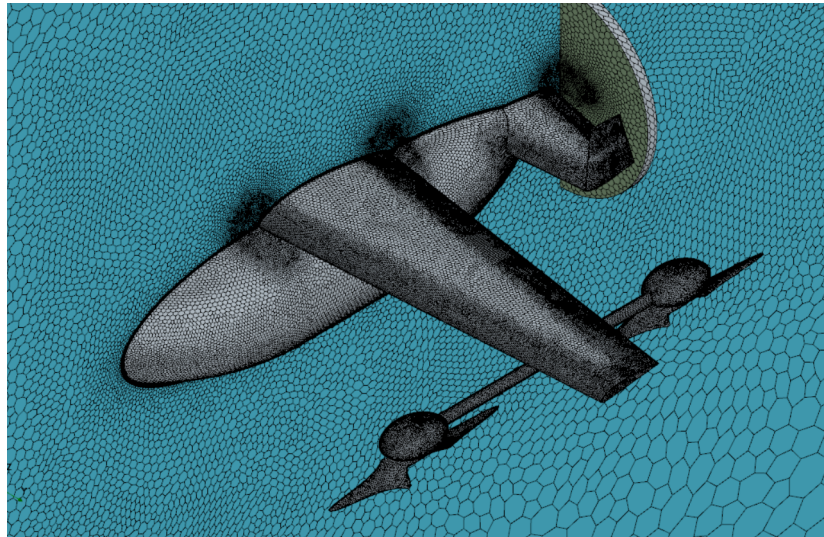


Figure 11: UAV Mesh for Star CCM+ Analysis

4.2.2 Lift and Drag

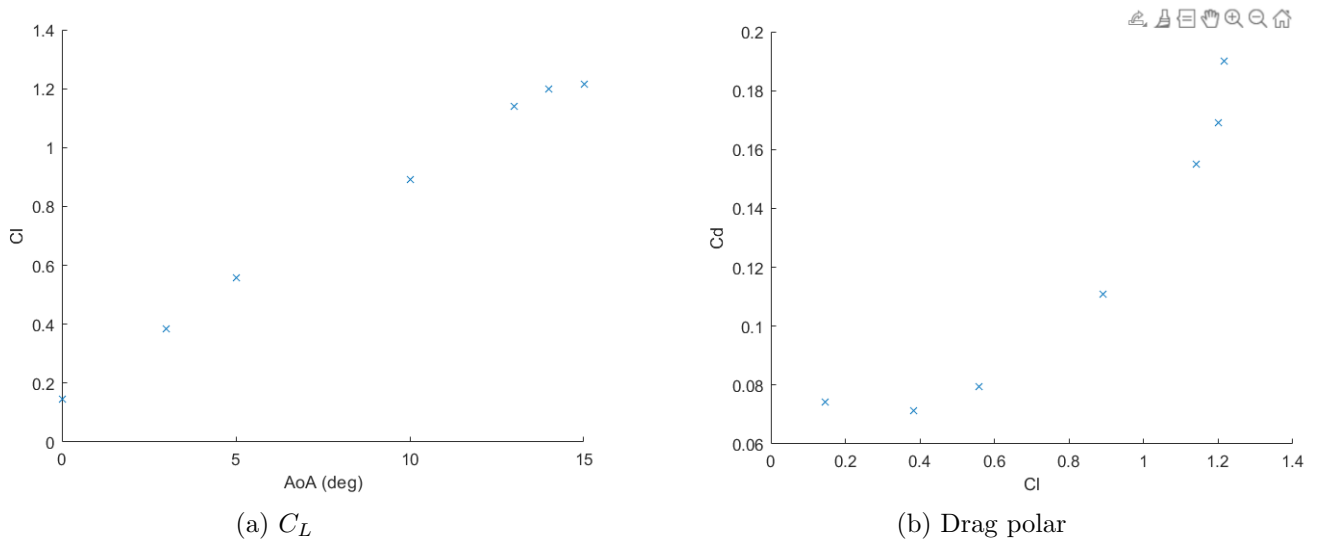


Figure 12: C_L Plot and Drag Polar of UAV

The lift and drag performance of the UAV from Star CCM+ results were plotted in Figure 12. Looking into the C_L plot, the UAV was able to achieve our desired C_{Lmax} of 1.25 for low enough stall speed for transition. However, it also showed that the UAV had a C_L of 0.146 at 0°freestream AoA, which was inadequate for steady level flight. The desired C_L at cruise was 0.347. Therefore a correction was made in the final design to account for that.

In terms of drag performance, the UAV had a L/D of 4.84 at cruise C_L and a L/D_{max} of 8.43 which happened at loiter. The overall drag of the aircraft was high due to the blunt and big fuselage compared to the wing, which was inevitable because of the large dimensions of life raft it was carrying. However, drag was not a major design concern because the our UAV was not designed for long flight and sustain.

4.2.3 Effect of Back Propeller

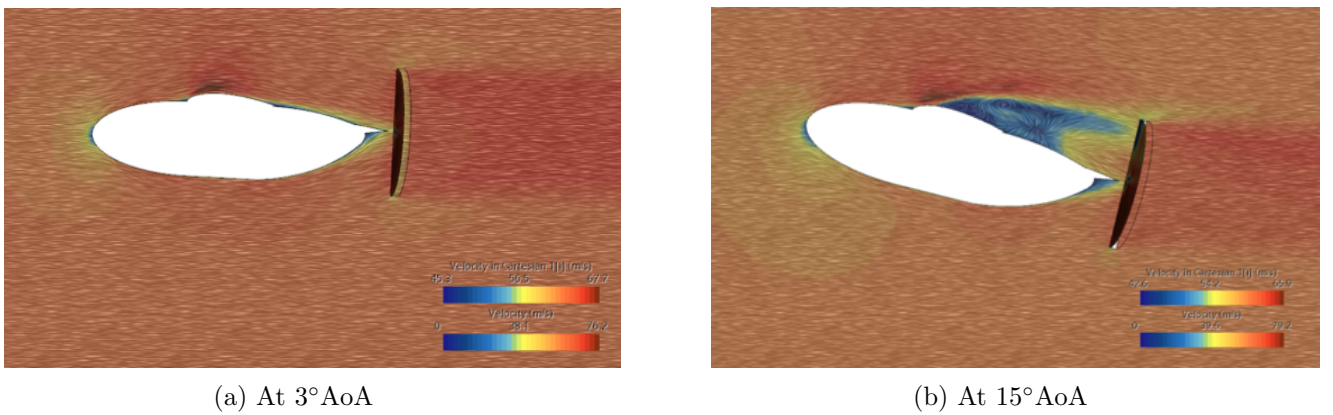


Figure 13: Velocity Plots Around the UAV

The pusher propeller on the back was found to have favourable effect on the overall aerodynamics of the UAV. As shown in Figure 13, the pusher prop helped to accelerate the flow and help them stay attached on the tapered back of the UAV. This has prevented separation and reduced the drag. At high AoA, it suppressed the wake of the main wing and allowed the tailplane to work at its designed condition.

5 Conclusions

In summary, writer's contributions to the aerodynamic design process of Fregata and the results have been outlined in this report. It was kickstarted by an initial wing planform design based on design requirements and values from the previous stages. It has generated a initial wing planform which was submitted to my colleague for refinement and optimisation. Following was a study on the nose design. Working closely with the structural team, it was designed to have an elliptical nose with length of 0.47 m and base radius of 0.176 m. Finally Star CCM+ analysis was done to validate the designs and give accurate aerodynamic performance data. It was also able to spot out some design flaws and driven the redesign of a more capable wing and the addition of aerodynamic fairings on the VTOL motors.

References

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A Iterative Matlab Code for Initial Wing Design

```
clear
clc
syms A1 A3 A5
```

```
S_ref = 51.5*9.81/832;
b = 2; %span
AR = b^2/S_ref;
Swept = 0;
Y = b/2; %semi-span
U_cruise = 58.3;
Cl_design = 51.5*9.81/(0.5*1.21*U_cruise^2*S_ref) % 3d Cl design
```

```
Cl_design = 0.4046
```

```
% airfoil data
a = 6.2743; % lift curve slope of selected airfoil, rad^-1
alpha_0lift = -4.3638; % zero lift angle of selected airfoil, degree
```

```
% tapered ratio
lambda = [0.5]; % suggested for unswept wing by raymer, can do iteration for lambda and twist later
twist = [-5:0.5:-2.5]; %degree
```

```
CL=zeros(length(lambda),length(twist));
CD_i=zeros(length(lambda),length(twist));
L_Di=zeros(length(lambda),length(twist));
e=zeros(length(lambda),length(twist));
for m=1:length(lambda)
    for n=1:length(twist)
        clear offset
% loop to get the settling angle right
Cl=0;
offset(1) = 1;
k = 1;
while abs (Cl-Cl_design) > 0.0005
    alpha_mean = rad2deg(Cl_design/a + deg2rad(alpha_0lift)); % AoA of geometric mean chord initial estimation
    c_root = 2*S_ref/(b*(1+lambda(m)));
    c_tip = lambda(m)*c_root;
    c = @(theta) c_root*(1-(1-lambda(m))*abs(cos(theta))); % theta in radian here
%aerodynamic mean chord
    c_mean = 2/3*c_root*(1+lambda(m)+lambda(m)^2)/(1+lambda(m));
    y_mean = (b/6)*(1+2*lambda(m))/(1+lambda(m));

%alpha calculation
    alpha_inf = 0; %freestream angle of attack
    alpha_root = alpha_mean-twist(n)/Y*y_mean+sum(offset); %degree
    alpha_tip = alpha_root+twist(n); %degree
    alpha = @(theta) alpha_inf + alpha_root + twist(n)* cos(theta) - alpha_0lift; % alpha in degree and theta in radian

%aerodynamic mean chord
    c_mean = 2/3*c_root*(1+lambda(m)+lambda(m)^2)/(1+lambda(m));
    y_mean = (b/6)*(1+2*lambda(m))/(1+lambda(m));

% lifting line theory, collocation method of 6 collocation points
    theta_1 = pi/7;
    theta_2 = 2*pi/7;
    theta_3 = 3*pi/7;

    mu_1 = a*c(theta_1)/(4*AR*c_mean);
    mu_2 = a*c(theta_2)/(4*AR*c_mean);
    mu_3 = a*c(theta_3)/(4*AR*c_mean);

    eqns = [mu_1*deg2rad(alpha(theta_1))*sin(theta_1) == A1*sin(theta_1)*(mu_1+sin(theta_1))+A3*sin(3*theta_1)*(3*mu_1+sin(theta_1))+A5*s
    Solution = solve(eqns,[A1 A3 A5]);

    Cl = vpa(pi*AR*Solution.A1);
    trackCl(k) = Cl ;

    if trackCl(1)<Cl_design
        if Cl<Cl_design
            k=k+1;
            offset(k) = offset(k-1);
        end

        if Cl>Cl_design
            offset(k) = offset(k)/2;
        end
    elseif trackCl(1)>Cl_design
        offset(1)=-1;
        if Cl>Cl_design
```

```

k=k+1;
offset(k) = offset(k-1);
end

if Cl>Cl_design
offset(k) = offset(k)/2;
end
end

CL(m,n) = double(round(vpa(pi*AR*Solution.A1),5));
CD_i(m,n) = double(round(vpa(pi*AR*Solution.A1^2*(1+3*(Solution.A3/Solution.A1)^2+5*(Solution.A5/Solution.A1)^2)),5));

L_Di(m,n) = round(CL(m,n)/CD_i(m,n),4);
e(m,n) = double(round(vpa(1/(1+3*(Solution.A3/Solution.A1)^2+5*(Solution.A5/Solution.A1)^2)),5));% span efficiency factor
end

[r,c]=find(L_Di==max(L_Di,[], 'all'));
CL(r(1),c(1))

```

```
ans = 0.4043
```

```
CD_i(r(1),c(1))
```

```
ans = 0.0080
```

```
L_Di(r(1),c(1))
```

```
ans = 50.4152
```

```
e(r(1),c(1))
```

```
ans = 0.9851
```

```

for i=1:length(r)
lambda(r(i))
twist(c(i))
end

```

```
ans = 0.5000
```

```
ans = -2.5000
```